Algorithm 1 Monte Carlo CP with $1-\alpha$ coverage guarantee for $m=1$ and $1-2\alpha$ for $m \geq 2$.

Input: Calibration examples $(X_i, \lambda_i)_{i\in[n]}$; test example X; confidence level α; number of samples m
Output: Prediction set $C(X)$ for test example

1. Sample m labels $(Y_i^j)_{j\in[m]}$ per calibration example $(X_i)_{i\in[n]}$ where $\mathbb{P}(Y_i^j = k) = \lambda_{ik}$.
2. Calibrate the threshold τ using

$$\tau = Q\left(\{E(X_i, Y_i^j)\}_{i\in[n], j\in[m]}; \frac{\lfloor \alpha m(n+1) \rfloor - m + 1}{mn}\right). \tag{10}$$

3. Return $C(X) = \{k \in [K] : E(X,k) \geq \tau\}$.

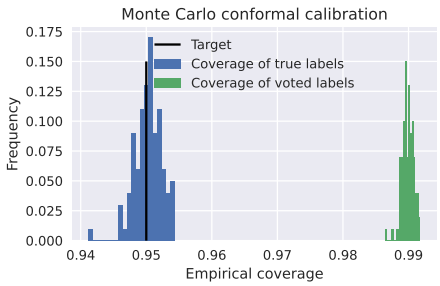

Figure 4: Monte Carlo CP with $m = 10$ sampled labels per calibration example (see Algorithm 1) applied to the toy dataset from Figure 3. We use the true posterior probabilities $\mathbb{P}(Y = y|X = x)$ as plausibilities λ_y to sample labels (see Figure 3, left). Again, we show coverage wrt. true or voted labels. In contrast to standard CP calibrated against voted labels, Monte Carlo CP clearly obtains the target coverage (black) of $1-\alpha = 95\%$ (on average) w.r.t. the true labels.

samples from $\mathbb{P}_{\text{agg}}(Y = y|X)$ as we only have access to one sample from B given X. We can only obtain exchangeable samples by repeating sampling from $\lambda \sim p(\lambda|B, X)$ and $\mathbb{P}(Y = y|\lambda) = \lambda_y$. Hence we cannot use concentration inequalities to obtain finite sample bounds for quantities such as $\mathbb{P}_{\text{agg}}(Y \in C(X)|X = x)$ for a given $C(X)$.

We will instead develop CP procedures returning prediction sets satisfying $\mathbb{P}_{\text{agg}}(Y \in C(X)) \geq 1-\alpha$, it is worth clarifying what this means when $\mathbb{P}_{\text{agg}}(Y|X)$ is ambiguous, i.e., not one-hot. With $\mathbb{P}_{\text{agg}} = \mathbb{P}^X \otimes \mathbb{P}_{\text{agg}}^{Y|X}$, we can rewrite the coverage guarantee as

$$\mathbb{P}_{\text{agg}}(Y \in C(X)) = \mathbb{E}_{X\sim\mathbb{P}^X}\left[\mathbb{E}_{Y\sim\mathbb{P}_{\text{agg}}(\cdot|X)}[\mathbb{I}[Y \in C(X)]]\right]. \tag{9}$$

It makes explicit that, coverage is marginal across examples *and* classes: in ambiguous cases, the prediction set $C(X)$ might cover only part of the classes with non-zero plausibility. We will refer to $\mathbb{P}_{\text{agg}}(Y \in C(X))$ defined in Equation (9) as **aggregated coverage** to emphasize that this is w.r.t. $\mathbb{P}_{\text{agg}}$. This is to contrast from **voted coverage** $\mathbb{P}_{\text{vote}}(Y \in C(X))$ which is w.r.t. $\mathbb{P}_{\text{vote}}$. To obtain prediction sets with aggregated coverage guarantees, we will assume access to a set of exchangeable calibration data $(X_i, \lambda_i)_{i\in[n]}$ of examples and corresponding plausibilities from $p(x, \lambda)$. To obtain this calibration data, we can simply rely on the original (exchangeable) calibration data $(X_i, B_i)_{i\in[n]}$ and then sample $\lambda_i \sim p(\cdot|X_i, B_i)$ where $\lambda_i = (\lambda_{i1}, ..., \lambda_{iK})$. In the examples given above, λ_i is given deterministically given (X_i, B_i) so $p(\lambda|x, b)$ is a delta-Dirac mass but, if a bootstrapping procedure is used to account for uncertainty in the annotation process, then it is not anymore.

3.2 Introduction to Monte Carlo conformal prediction

We propose Monte Carlo CP, a sampling-based approach to CP under ambiguous ground truth. Given the calibration examples $(X_i, \lambda_i)_{i\in[n]}$, we sample m labels Y_i^j for each X_i according to plausibilities λ_i, i.e. $\mathbb{P}(Y_i^j = k) = \lambda_{ik}$ and duplicate the corresponding inputs[4]. That is, we obtain $m \cdot n$ new calibration examples $(X_i, Y_i^j)_{i\in[n], j\in[m]}$ and then apply the conformal calibration outlined in Algorithm 1. As shown in Figure 4, Monte Carlo CP overcomes the coverage gap illustrated on our toy dataset in Figure 3.

[4]An alternative valid procedure would sample m plausabilities $\lambda_i^j \sim p(\cdot|X_i, B_i)$ for each X_i and then Y_i^j such that $\mathbb{P}(Y_i^j = k) = \lambda_{ik}^j$. When $p(\cdot|X_i, B_i)$ is a delta-mass distribution for all i, this coincides with the procedure described previously.

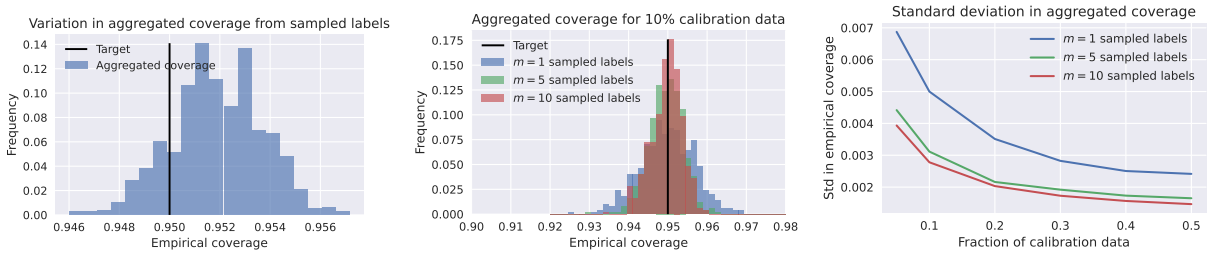

Figure 5: Left: Empirical aggregated coverage for Monte Carlo CP, $m = 1$, with a fixed calibration/test split but different randomly sampled labels Y_i^j, cf. Algorithm 1. The approach obtains coverage $1-\alpha$ across calibration/test splits but overestimates coverage slightly for this particular split. However, in Monte Carlo CP, coverage is marginal across not only test and calibration examples but also the sampled labels during calibration, as shown in this histogram. Middle and right: Instead fixing the sampled labels for different m and plotting variation in coverage across random calibration/test split shows that aggregated coverage is empirically close to $1-\alpha$ even for $m > 1$. Large m generally reduces the variability in coverage observed across calibration/test splits, especially for small calibration sets (we consider 5% to 50% calibration data).

The aggregated coverage guarantees we will provide for Algorithm 1 are marginal over the sampled labels $(Y_i^j)_{i\in[n],j\in[m]}$. This is illustrated in Figure 5 (left) on the toy dataset of Section 2.2 using a fixed calibration/test split but multiple samples of $(Y_i^j)_{i\in[n],j\in[m]}$. The empirical aggregated coverage across calibration/test splits is $1-\alpha$ (see Figure 5 (left)) but the variability across such splits is high if the entropy of $\mathbb{P}_{\text{agg}}^{Y|X}$ is high for many calibration data. This variability can be reduced by increasing m (see Figure 5, right), this is especially beneficial in the low calibration data regime.

We now discuss the theoretical coverage properties of Algorithm 1. Recall that we have assumed in Section 3.1 that the joint distribution of $(X_i, B_i)_{i\in[n]}$ and a test data (X, B) is exchangeable. This implies that the joint distribution of $(X_i, Y_i^j)_{i\in[n]}$ and (X, Y) is exchangeable for any $j \in [m]$. For $m = 1$, it is clear that this approach boils down to standard split CP applied to calibration data $(X_i, Y_i^1) \sim \mathbb{P}_{\text{agg}}$ and thus $\mathbb{P}_{\text{agg}}(Y \in C(X)) \geq 1-\alpha$ follows directly. For $m \geq 2$, the calibration examples include m repetitions of each X_i and exchangeability with the test example X, typically used to establish the validity of CP, is not satisfied anymore. Nevertheless, as mentioned earlier, we empirically observe an empirical coverage close to $1-\alpha$ on average in Figure 5 (middle).

In the following, Section 3.3, we focus on establishing rigorous coverage guarantees for Monte Carlo CP when $m \geq 2$, showing that we can establish $\mathbb{P}_{\text{agg}}(Y \in C(X)) \geq 1-2\alpha$. This is akin to the observation in (Barber et al., 2021) that jackknife+, cross-conformal or out-of-bag CP consistently obtain empirical coverage $1-\alpha$ while they only guarantee $1-2\alpha$. In applications where rigorous coverage guarantees better than $1-2\alpha$ are necessary, we show in Section 3.4 that this can be achieved by developing an alternative Monte Carlo CP procedure presented in Algorithm 2 which relies on an additional split of the calibration examples.

3.3 Coverage $1-2\alpha$ by averaging p-values

In order to establish the coverage guarantee $\mathbb{P}_{\text{agg}}(Y \in C(X)) \geq 1-2\alpha$ when $m \geq 2$ for Algorithm 1, let us introduce for $j \in [m]$:

$$\rho_Y^j = \frac{\sum_{i=1}^n \mathbb{I}[E(X_i, Y_i^j) \leq E(X, Y)] + 1}{n+1}. \tag{11}$$

The random variables ρ_Y^j are p-values, i.e. $\mathbb{P}(\rho_Y^j \leq \alpha) \leq \alpha$, since the scores $\{E(X_i, Y_i^j)\}_{i\in[n]}$ and $\{E(X, Y)\}$ are exchangeable for fixed $j \in [m]$. We can now average these quantities over $j \in [m]$ to obtain

$$\bar{\rho}_Y = \frac{1}{m}\sum_{j=1}^m \rho_Y^j = \frac{\sum_{j=1}^m \left(\sum_{i=1}^n \mathbb{I}[E(X_i, Y_i^j) \leq E(X, Y)] + 1\right)}{m(n+1)}. \tag{12}$$

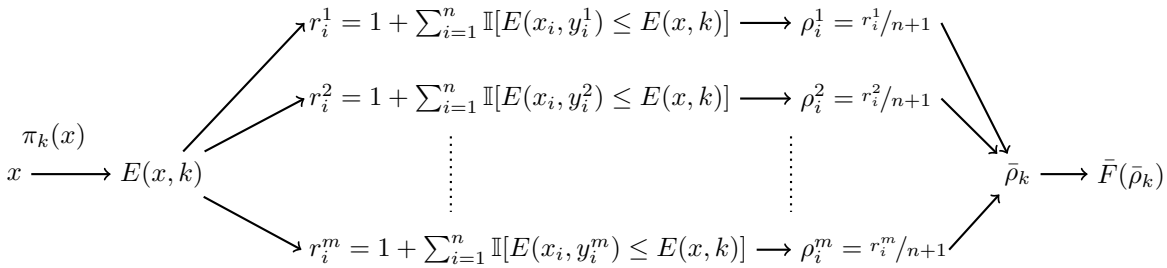

Figure 6: The m label samples result in $(\rho_k^j)_{j\in[m]}$ for the test example X, $(\rho_Y^j)_{j\in[m]}$ being p-values. We can average ρ_k^j over j and threshold the resulting average $\bar{\rho}_k$ at level α to obtain the prediction set, this is equivalent to Algorithm 1 as explained in Section 3.3. Alternatively we can combine the p-values ρ_Y^j using the ECDF approach described in Section 3.4; see Algorithm 2 for a detailed computational description.

As $\bar{\rho}_Y$ is an average of (dependent) p-values, it follows from standard results (Rüschendorf, 1982; Meng, 1994) that $\mathbb{P}(2\bar{\rho}_Y \leq 2\alpha) \leq 2\alpha$, equivalently $\mathbb{P}(\bar{\rho}_Y > \alpha) \geq 1 - 2\alpha$. Hence we have $\mathbb{P}(Y \in C(X)) \geq 1 - 2\alpha$ for $C(X) = \{y \in [K] : \bar{\rho}_y > \alpha\}$. Nevertheless, as discussed previously, we obtain empirical coverage close to $1 - \alpha$, cf. Figure 5 (middle), see also the discussion in (Barber et al., 2021, Section 4). This approach is illustrated in Figure 6. Note that in the context of aggregating m different models, Linusson et al. (2017) also considered similar types of averaging.

Practically, we do not have to compute and actually average ρ_y^j for $j \in [m]$ and $y \in [K]$ to determine $C(X) = \{y \in [K] : \bar{\rho}_y > \alpha\}$. As in Section 2.1, we can reformulate this prediction set as $C(X) = \{k \in [K] : E(X,k) \geq \tau\}$ as in Equation (2) where τ is the $\lfloor \alpha m(n+1) \rfloor - m + 1$ smallest element of $\{E(X_i, Y_i^j)\}_{i\in[n], j\in[m]}$, equivalently τ is obtained by computing

$$\tau = Q\left(\{E(X_i, Y_i^j)\}_{i\in[n], j\in[m]}; \frac{\lfloor \alpha m(n+1) \rfloor - m + 1}{mn}\right). \tag{13}$$

and confidence sets are constructed as $C(X) = \{k \in [K] : E(X,k) \geq \tau\}$ for τ given in Equation (10). This is established in Section B.2 of Appendix B.

3.4 Beyond coverage $1 - 2\alpha$

The $1-2\alpha$ coverage guarantee from Monte Carlo CP arises from the fact that we use a standard result about average of p-values (Rüschendorf, 1982; Meng, 1994). When the p-values are independent, a few techniques have been proposed to obtain a $1 - \alpha$ coverage; see e.g. (Cinar & Viechtbauer, 2022) for a comprehensive review. However, in our case, the p-values $(\rho_Y^j)_{j\in[m]}$ we want to combine are strongly dependent as they use the same calibration examples X_i and then rely on different pseudo-labels Y_i^j from the same distribution (given by λ_i). In this setting, many standard methods yield overly conservative results.

We follow here a method that directly estimates the cumulative distribution function (CDF) of the combined, e.g., averaged p-values (Balasubramanian et al., 2015; Toccaceli & Gammerman, 2019; Toccaceli, 2019). Let $\bar{\rho}_Y = {}^1/_m \sum_{j=1}^m \rho_Y^j$ be the averaged p-values. As shown in Figure 7 (left), these averaged p-values will not be uniformly distributed. However, if F denotes the CDF of $\bar{\rho}_Y$, then $\rho_Y = F(\bar{\rho}_Y)$ is a p-value[5] and thus the prediction set $C(X) = \{y \in [K] : \rho_y > \alpha\}$ will obtain coverage $1 - \alpha$. The true CDF is unknown but we can split the original calibration examples into $X_1, \ldots, X_l$ and $X_{l+1}, \ldots, X_n$ and used the second split to obtained an empirical estimate $\bar{F}$ of F. The procedure is described in Algorithm 2.

As $\bar{F}$ is not the true CDF F but an empirical CDF (ECDF) estimate, $C(X) = \{y \in [K] : \bar{F}(\bar{\rho}_y) > \alpha\}$ would only provide an approximate coverage guarantee at level $1 - \alpha$. However, if the original calibration examples $X_{l+1}, \ldots, X_n$ are i.i.d., then the Dvoretzky–Kiefer–Wolfowitz inequality (see e.g. (Wasserman,

[5] We have $\mathbb{P}(F(\bar{\rho}_Y) \leq f) = \mathbb{P}(F(F^{-1}(U) \leq f)$ for U a uniform random variable on $[0,1]$ for $F^{-1}(f) = \inf\{\rho \in \mathbb{R} : F(\rho) \geq f\}$. However $F(F^{-1}(U)) \geq U$ so $\mathbb{P}(F(F^{-1}(U) \leq f) \leq \mathbb{P}(U \leq f) = f$. Hence, we have $\mathbb{P}(\rho_Y \leq \alpha) \leq \alpha$.